JNTUA R23 REGULATIONS

ENVIRONMENTAL ENGINEERING LAB

(23A01603P)

Laboratory Manual with Detailed Inputs, Procedures, Calculations and Outputs

Programme	Year / Semester	Course code	L	T	P	C
B.Tech. Civil Engineering	III B.Tech – II Semester	23A01603P	0	0	3	1.5

Course Objectives — to make the student:

1. **Understand** the principles and methods of water and wastewater sampling and preservation.
2. **Perform** standard laboratory tests to determine water quality parameters.
3. **Analyze** wastewater characteristics and assess pollution levels.
4. **Evaluate** the effectiveness of treatment processes using chemical and biological tests.
5. **Develop** hands-on skills in advanced laboratory techniques for environmental monitoring.

Course Outcomes — on completion, students will be able to:

CO	Outcome	Covered in
CO1	Apply appropriate sampling and preservation techniques for water and wastewater	II.1
CO2	Measure physical and chemical parameters such as turbidity, conductivity and chlorine content	II.2 – II.5, II.7
CO3	Analyze key water and wastewater quality indicators like BOD, COD and TKN	III.3 – III.6
CO4	Assess the efficiency of water and wastewater treatment processes through laboratory tests	II.6, II.7, III.1, III.2
CO5	Perform microbiological analysis for coliform detection and sludge characterization	III.2, III.7

The syllabus sheet numbers the outcomes 2–6 under the heading of five outcomes; here they are numbered CO1–CO5 to match the CO–PO articulation matrix.

Note: a minimum of **10 out of the 14 experiments** listed below must be carried out.

How to read each experiment

Every experiment follows the layout **1 Aim → 2 Inputs** (apparatus, reagents, quantities) **→ 3 Procedure → 4 Observations → 5 Calculations → 6 Outputs → 7 Precautions**. Observation tables hold *illustrative sample data*, calculated consistently, to show the full calculation chain and the form of the final output; students must substitute their own readings. Limits quoted are from IS 10500:2012 (drinking water), CPCB/EPA Schedule VI effluent standards and APHA *Standard Methods*; verify against current editions before submission.

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No.	Experiment	Method / Reference
	II. ANALYSIS OF WATER SAMPLE	
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II.2	Measurement of electrical conductivity and turbidity	IS 3025 (Pt 14, Pt 10); APHA 2510, 2130
II.3	Determination of fluoride in water (spectrophotometric SPADNS / ISE)	IS 3025 (Part 60); APHA 4500-F
II.4	Determination of iron in water (Demo)	IS 3025 (Part 53); APHA 3500-Fe B
II.5	Determination of sulphate in water	IS 3025 (Part 24); APHA 4500-SO_4 E
II.6	Determination of optimum coagulant dosage by Jar test apparatus	IS 3025 / CPHEEO Manual
II.7	Available chlorine in bleaching powder and residual chlorine in water	IS 1065; APHA 4500-Cl B
	III. ANALYSIS OF WASTEWATER SAMPLE	
III.1	Estimation of suspended, volatile and fixed solids	IS 3025 (Part 17, 18); APHA 2540
III.2	Determination of Sludge Volume Index (SVI) in wastewater	APHA 2710 D
III.3	Determination of dissolved oxygen (Winkler, azide modification)	IS 3025 (Part 38); APHA 4500-O C
III.4	Estimation of B.O.D.	IS 3025 (Part 44); APHA 5210 B
III.5	Estimation of C.O.D.	IS 3025 (Part 58); APHA 5220 B
III.6	Determination of TKN and ammonia nitrogen in wastewater	IS 3025 (Part 34); APHA 4500-N_{org}, NH_3
III.7	Determination of total and faecal coliform (Demonstration only)	IS 1622; APHA 9221 (MPN)
A	Appendix — Summary table, quality standards, safety, text & reference books	—

Experiment II.1: Sampling and Preservation Methods for Water and Wastewater (Demonstration)

Course Outcome: CO1 | **Reference standard:** IS 3025 (Part 1); APHA 1060 (collection and preservation)

1. Aim

To learn the correct methods of collecting representative grab and composite samples of water and wastewater and of preserving them until analysis.

2. INPUTS — Apparatus, Materials & Quantities

Item	Specification / Quantity
Sample containers	Clean polyethylene / borosilicate glass bottles (1–2 L), 300 mL BOD bottles, sterile bottles for bacteriology
Sampling devices	Sampling bucket / dipper on rope, Kemmerer or Van Dorn sampler, depth sampler, composite sampler (optional)
Preservatives	Conc. H_2SO_4, conc. HNO_3, $MnSO_4$, alkali-iodide-azide, 10% sodium thiosulphate
Field equipment	Ice box with ice packs, thermometer, pH paper, labels, marker, gloves, field data sheet, chain-of-custody form
Sample volume	Physico-chemical: 2 L; BOD/COD: 1 L; bacteriological: 250 mL; heavy metals: 1 L

3. Procedure

1. Select the sampling point (well-mixed, representative; for taps, flush 2–3 min; for rivers, mid-stream, 15–30 cm below surface, facing upstream).
2. Rinse the container 2–3 times with the sample (except sterile bottles and bottles containing preservative).
3. Collect a **grab sample** for unstable parameters (DO, pH, residual chlorine, temperature) and a **composite sample** (time- or flow-proportioned) for wastewater strength.
4. Fill BOD/DO bottles completely with no air bubbles; for bacteriology, leave a 2.5 cm air space and do not touch the neck or cap.
5. Add the required preservative immediately, label (ID, date, time, place, collector, preservative, tests) and place in an ice box at 4 °C.
6. Record on-site readings, transport to the laboratory within the holding time, and record in the chain-of-custody form.

4. Observations — Preservation table (reference) and flow-proportioned composite sample

Sample readings are illustrative; students must replace them with their own laboratory readings.

Parameter	Container	Preservation	Max. holding time
pH, EC, turbidity	Plastic / glass	Cool to 4 °C; no air space	pH: analyse at once; turbidity 24 h; EC 28 d
Fluoride	Plastic (not glass)	None required	28 days
Total iron	Acid-washed plastic / glass	HNO_3 to pH < 2	6 months
Sulphate	Plastic / glass	Cool to 4 °C	28 days
Dissolved oxygen	300 mL BOD bottle	Fix on site: 2 mL $MnSO_4$ + 2 mL alkali-iodide-azide; store dark	8 h (after fixing)
BOD	Plastic / glass, filled full	Cool to 4 °C, dark	6 h preferred; max 48 h
COD	Glass	H_2SO_4 to pH < 2; 4 °C	28 days
TKN / ammonia-N	Plastic / glass	H_2SO_4 to pH < 2; 4 °C	28 days
Solids (TS, TSS, VSS)	Plastic / glass	Cool to 4 °C	7 days
Residual chlorine	Glass, amber	None — analyse immediately	Immediately (≤ 15 min)
Coliforms	Sterile glass; $Na_2S_2O_3$ (0.1 mL of 10%/125 mL) if chlorinated	Cool to < 10 °C, dark	6 h (max 24 h)

Hour	Flow Q (L/s)	Aliquot $V_i = V \times Q_i / \Sigma Q$ (mL)
1	10	370
2	14	519

Hour	Flow Q (L/s)	Aliquot $V_i = V \times Q_i / \Sigma Q$ (mL)
3	18	667
4	12	444
Total	54	2000

5. Calculations

Flow-proportioned composite: aliquot volume $V_i = V_{total} \times Q_i \div \Sigma Q_i$

Hour 1: $V_1 = 2000 \times 10 \div 54 = 370$ mL; Hour 3: $V_3 = 2000 \times 18 \div 54 = 667$ mL

Time-proportioned (equal aliquots) composite is used when flow is nearly constant: $V_i = V_{total} \div n$

6. OUTPUTS — Results from the sample data

Output parameter	Value	Remark
Composite sample volume	2000 mL	Aliquots 370, 519, 667, 444 mL
Preservation record	Labelled & iced (4 °C)	As per the table above
Sample validity	Within holding time	See table (e.g., BOD 6–48 h)

Result / Acceptance criteria

A representative, correctly preserved sample was obtained; the **preservation method and holding time** for each parameter are as tabulated. Improper sampling is the largest source of error in water-quality analysis.

7. Precautions

- Never touch the inside of the container or cap; avoid contamination from hands, gloves and sampling gear.
- Do not use glass for fluoride; do not use plastic for oil & grease.
- Handle acids with care while preserving samples in the field; wear gloves and goggles.

Experiment II.2: Measurement of Electrical Conductivity and Turbidity

Course Outcome: CO2 | **Reference standard:** IS 3025 (Part 14) – conductivity; IS 3025 (Part 10) – turbidity; APHA 2510 B, 2130 B

1. Aim

To determine the electrical conductivity (a measure of dissolved ionic strength) and the turbidity (suspended matter) of given water samples, and to compare them with drinking-water limits.

2. INPUTS — Apparatus, Materials & Quantities

Item	Specification / Quantity
Conductivity meter	Digital meter with cell (cell constant ≈ 1.0 cm^{-1}) and temperature probe (ATC)
Standard solution	0.01 M KCl (0.7455 g KCl, dried at 180 °C, in 1 L conductivity water) → κ = 1413 µS/cm at 25 °C
Nephelometer	Turbidity meter reading in NTU, with clean matched sample cells
Turbidity standards	Formazin stock 4000 NTU (1 g hydrazine sulphate + 10 g hexamethylenetetramine per litre) diluted to 0, 10, 100 NTU, or sealed Gelex standards
Others	Distilled water (EC < 2 µS/cm), beakers, tissue paper, thermometer; samples: tap, bore-well, pond water (250 mL each)

3. Procedure

1. **Conductivity:** rinse the cell with distilled water and then with 0.01 M KCl; calibrate at 25 °C so the meter reads 1413 µS/cm.
2. Rinse the cell with the sample, immerse it fully (no air bubbles), stir gently, wait for a stable reading and record the conductivity and temperature.
3. Convert the reading to 25 °C; estimate TDS ≈ k × EC_{25} (k = 0.55–0.75; 0.65 for natural waters).
4. **Turbidity:** switch on the nephelometer to warm up; calibrate using 0 NTU (distilled water) and formazin standards.
5. Shake the sample gently (avoid bubbles), fill the cell to the mark, wipe the outside with tissue, insert and read NTU. If > 40 NTU, dilute with turbidity-free water and multiply by the dilution factor.

4. Observations — Conductivity and turbidity readings

Sample readings are illustrative; students must replace them with their own laboratory readings.

Sample	Temp. T (°C)	Measured EC (µS/cm)	EC at 25 °C (µS/cm)	Est. TDS = 0.65 EC (mg/L)	Turbidity (NTU)
Tap water	28	395	373	242	0.8
Bore well	27	1020	981	638	2.4
Pond water	26	610	598	389	18.6

5. Calculations

Temperature correction: $EC_{25} = EC_T \div [1 + 0.02\,(T - 25)]$

Bore well: $EC_{25} = 1020 \div [1 + 0.02 \times 2] = 1020 \div 1.04 = 981$ µS/cm

TDS (mg/L) ≈ 0.65 × EC_{25} = 0.65 × 981 = 638 mg/L

Cell constant $K = \kappa_{standard} \div G_{measured}$; $\kappa = K \times G$

6. OUTPUTS — Results from the sample data

Output parameter	Value	Remark
Tap water	EC 373 µS/cm; 0.8 NTU	TDS ≈ 242 mg/L ≤ 500; turbidity ≤ 1 NTU → **acceptable**
Bore well	EC 981 µS/cm; 2.4 NTU	TDS ≈ 637 mg/L (> 500, < 2000); turbidity 1–5 NTU → **permissible** only if no alternate source
Pond water	EC 598 µS/cm; 18.6 NTU	TDS ≈ 389 mg/L; turbidity > 5 NTU → **needs treatment** (coagulation – Exp. II.6)

Result / Acceptance criteria

IS 10500:2012 limits: **turbidity 1 NTU** (acceptable) – 5 NTU (permissible); **TDS 500 mg/L** (acceptable) – 2000 mg/L (permissible). Conductivity and turbidity of the samples are as tabulated.

7. Precautions

- Do not touch the electrodes; keep the cell clean and always store it as per the manufacturer’s instructions.
- Remove air bubbles from cells; scratched/dirty cells give wrong NTU readings.
- Report conductivity always with the reference temperature (25 °C).

Experiment II.3: Determination of Fluoride in Water (SPADNS Spectrophotometric Method / ISE)

Course Outcome: CO2 | **Reference standard:** IS 3025 (Part 60); APHA 4500-F D (SPADNS), 4500-F C (ISE)

1. Aim

To determine the fluoride concentration in water by the colorimetric SPADNS method (and note the ion-selective electrode alternative) and compare it with the drinking-water limit.

2. INPUTS — Apparatus, Materials & Quantities

Item	Specification / Quantity
Spectrophotometer	Visible, 570 nm, 1 cm cells
Stock fluoride	100 mg F^-/L: 221.0 mg anhydrous NaF (dried at 110 °C) in 1 L distilled water
Working standard	10 mg F^-/L: dilute 100 mL of stock to 1 L
SPADNS reagent	958 mg SPADNS in 500 mL distilled water
Zirconyl-acid reagent	133 mg $ZrOCl_2 \cdot 8H_2O$ in 25 mL water + 350 mL conc. HCl diluted to 500 mL; mix equal volumes of the two reagents for use
Reference solution	10 mL SPADNS + 10 mL dil. HCl (7 mL conc. HCl + 10 mL water) made to 100 mL
Others	50 mL Nessler tubes / volumetric flasks, pipettes, water sample (100 mL, filtered)

3. Procedure

1. Prepare standards of 0, 0.2, 0.4, 0.6, 0.8, 1.0 and 1.2 mg/L by taking 0, 1, 2, 3, 4, 5, 6 mL of working standard into 50 mL flasks and diluting to 50 mL.
2. Add 10 mL of mixed acid-zirconyl-SPADNS reagent to each standard, mix well and hold for 10 min (same temperature for all).
3. Set the spectrophotometer to zero absorbance with the reference solution at 570 nm; read the absorbance of every standard.
4. Take 50 mL of the sample (dilute if necessary), add 10 mL reagent, mix, and read absorbance after the same time interval.
5. Plot absorbance vs fluoride (a straight line of *negative* slope — colour fades as fluoride increases) and read the sample concentration.
6. **ISE alternative:** mix 25 mL sample with 25 mL TISAB, calibrate the F^- electrode with 0.1, 1, 10 mg/L standards (slope ≈ −54 to −60 mV/decade) and read the sample directly.

4. Observations — Calibration data (λ = 570 nm)

Sample readings are illustrative; students must replace them with their own laboratory readings.

Fluoride (mg/L)	0.0	0.2	0.4	0.6	0.8	1.0	1.2
Absorbance	0.640	0.600	0.560	0.520	0.480	0.440	0.400

Sample	Absorbance	Dilution factor
Bore-well water	0.538	1 (none)

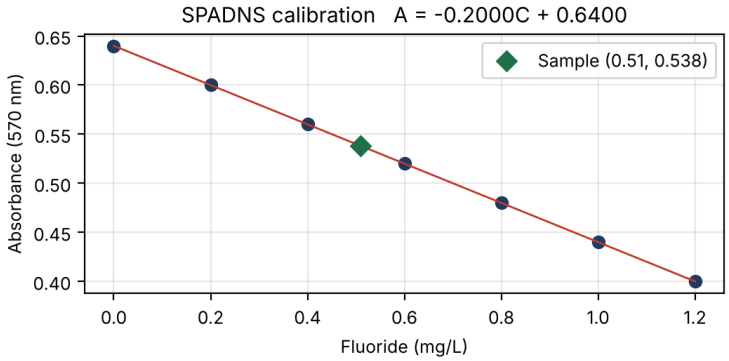


Fig. II.3 Fluoride calibration curve (SPADNS)

5. Calculations

Calibration line (least squares): A = -0.200 × C + 0.640

Sample: C = (A − 0.640) ÷ (-0.200) = (0.538 − 0.640) ÷ (−0.200) = 0.51 mg/L

Fluoride (mg/L) = C_{read} × Dilution factor

6. OUTPUTS — Results from the sample data

Output parameter	Value	Remark
Fluoride in sample	0.51 mg/L	IS 10500: 1.0 acceptable; 1.5 permissible
Status	Within limit	< 1.0 mg/L – safe against fluorosis; optimal 0.7–1.0
Linearity of curve	R^2 = 1.000 (sample data)	Use only the range of the standards

Result / Acceptance criteria

Fluoride concentration of the given water = **0.51 mg/L**, which is below the IS 10500 acceptable limit of 1.0 mg/L.

7. Precautions

- Standards and samples must be at the same temperature; add reagent identically and read at the same time.
- Interferences: alkalinity, chlorine, aluminium, iron, phosphate — remove chlorine with sodium arsenite (or thiosulphate).
- Use plastic ware for fluoride; SPADNS contains hazardous chemicals, handle with gloves.

Experiment II.4: Determination of Iron in Water (Demonstration) — Phenanthroline Method

Course Outcome: CO2 | **Reference standard:** IS 3025 (Part 53); APHA 3500-Fe B

1. Aim

To determine the total iron in water colorimetrically using 1,10-phenanthroline, which forms an orange-red complex with Fe^{2+}.

2. INPUTS — Apparatus, Materials & Quantities

Item	Specification / Quantity
Spectrophotometer / photometer	510 nm, 1 cm cells (or green filter)
Stock iron	Dissolve 0.7022 g $Fe(NH_4)_2(SO_4)_2{\cdot}6H_2O$ in 50 mL water + 20 mL conc. H_2SO_4, $KMnO_4$ drop to pink; dilute to 1 L (1 mL = 100 µg Fe); working: 10 mg/L
Reagents	Conc. HCl; hydroxylamine hydrochloride (10%); phenanthroline (0.1% in water, warm); ammonium acetate buffer (250 g $NH_4C_2H_3O_2$ + 700 mL acetic acid in 1 L)
Glassware	100 mL conical flasks, 50 mL volumetric flasks, pipettes, hot plate; all acid-washed
Sample	50 mL (well-mixed, acidified at collection)

3. Procedure

1. Prepare 50 mL standards containing 0, 0.2, 0.4, 0.6, 0.8 and 1.0 mg Fe/L (0–5 mL of 10 mg/L working standard).
2. To 50 mL of sample (and each standard) add 2 mL conc. HCl and 1 mL hydroxylamine solution; boil gently until volume reduces to 15–20 mL (to dissolve iron and reduce Fe^{3+} to Fe^{2+}). Cool.
3. Transfer to a 50 mL flask; add 10 mL ammonium acetate buffer and 2 mL phenanthroline solution; make up to the mark.
4. Wait 10–15 min for full colour development; read the absorbance at 510 nm against the blank (0 mg/L).
5. Plot the calibration curve and find the iron in the sample from its absorbance.

4. Observations — Calibration data (λ = 510 nm)

Sample readings are illustrative; students must replace them with their own laboratory readings.

Iron (mg/L)	0.0	0.2	0.4	0.6	0.8	1.0
Absorbance	0.000	0.040	0.080	0.119	0.159	0.199

Sample	Absorbance	Sample volume (mL)	Dilution
Bore-well water	0.052	50	None

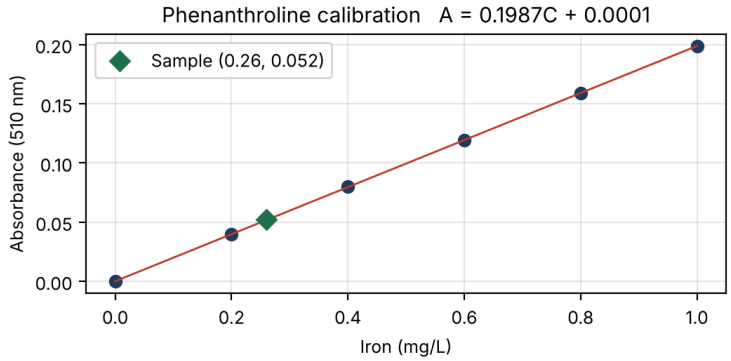


Fig. II.4 Iron calibration curve

5. Calculations

Calibration line: A = 0.199 × C + 0.0001

Sample: C = (0.052 − 0.0001) ÷ 0.199 = 0.26 mg/L

Iron (mg/L) = C_{read} × (50 ÷ mL of sample taken) × dilution factor

Check: molar absorptivity of Fe-phenanthroline ≈ 11 100 L mol^{-1} cm^{-1} → A ≈ 0.20 per mg/L in 1 cm cell

6. OUTPUTS — Results from the sample data

Output parameter	Value	Remark
Total iron in sample	0.26 mg/L	IS 10500: 0.3 mg/L (acceptable = permissible)
Status	Within limit (marginal)	> 0.3 mg/L gives taste, staining; treat by aeration + filtration

Result / Acceptance criteria

Total iron in the water = **0.26 mg/L** (limit 0.3 mg/L), so the sample is acceptable but close to the limit.

7. Precautions

- Acid-wash all glassware; avoid contact with metallic objects.
- Ensure complete reduction of Fe^{3+} (boil with hydroxylamine); add the reagents in the stated order.
- Strong acids and phenanthroline are toxic — wear gloves and goggles.

Experiment II.5: Determination of Sulphate in Water (Turbidimetric Method)

Course Outcome: CO2 | **Reference standard:** IS 3025 (Part 24); APHA 4500-SO_4^{2-} E

1. Aim

To determine the sulphate concentration of water by precipitating it as barium sulphate and measuring the resulting turbidity.

2. INPUTS — Apparatus, Materials & Quantities

Item	Specification / Quantity
Spectrophotometer / nephelometer	420 nm (or turbidity meter), 1 cm cells
Magnetic stirrer	Constant speed, with 25 mm stirring bar
Conditioning reagent	30 mL conc. HCl + 300 mL distilled water + 100 mL 95% ethanol (or isopropanol) + 75 g NaCl + 50 mL glycerol; make to 1 L
Barium chloride	$BaCl_2 \cdot 2H_2O$ crystals, 20–30 mesh
Standard sulphate	Stock 100 mg SO_4^{2-}/L: 147.9 mg anhydrous Na_2SO_4 per litre
Others	250 mL beakers, 100 mL volumetric flasks, 20 mL sample aliquot (diluted to 100 mL)

3. Procedure

1. Prepare standards 0–40 mg/L (0, 5, 10, 15, 20, 25, 30, 40) in 100 mL volumetric flasks from the stock solution.
2. Pipette the sample (20 mL here) into 100 mL flask and make up to the mark (dilution factor = 5); transfer to a 250 mL beaker.
3. Add 5 mL conditioning reagent, mix on the stirrer; while stirring add a spoonful (≈ 0.2–0.3 g) of $BaCl_2$ crystals and stir for exactly 60 s at constant speed.
4. Immediately pour into the cell and measure the absorbance (or NTU) at 30 s intervals for 4 min; take the maximum reading (usually within 4 min).
5. Prepare a sample blank (without $BaCl_2$) and subtract colour/turbidity. Plot the calibration curve and read the sulphate concentration.

4. Observations — Calibration data (λ = 420 nm)

Sample readings are illustrative; students must replace them with their own laboratory readings.

SO_4^{2-} (mg/L)	0	5	10	15	20	25	30	40
Absorbance	0.002	0.057	0.112	0.167	0.222	0.277	0.332	0.442

Sample	Aliquot (mL)	Final volume (mL)	Dilution factor	Absorbance (net)
Bore-well water	20	100	5	0.200

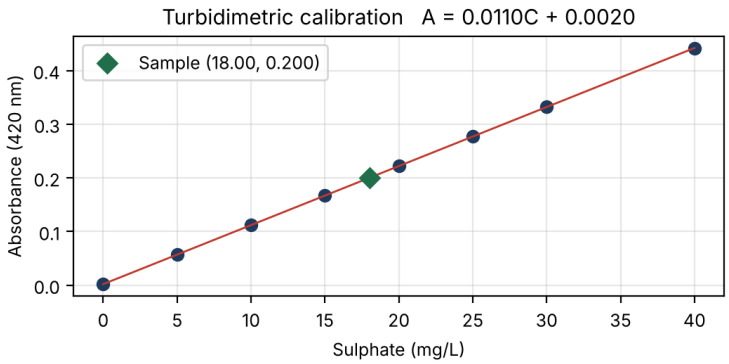


Fig. II.5 Sulphate calibration curve

5. Calculations

Calibration line: A = 0.0110 × C + 0.0020

Concentration in diluted sample = (0.200 − 0.0020) ÷ 0.0110 = 18.0 mg/L

Sulphate in original sample = 18.0 × 5 = 90 mg/L

SO_4^{2-} (mg/L) = C_{read} × (100 ÷ mL sample)

6. OUTPUTS — Results from the sample data

Output parameter	Value	Remark
Sulphate in sample	90 mg/L	IS 10500: 200 acceptable; 400 permissible
Status	Within acceptable limit	High sulphate (> 250 mg/L) causes laxative effect, corrosion of concrete

Result / Acceptance criteria

Sulphate concentration of the water = **90 mg/L** (< 200 mg/L) — acceptable for drinking and for use in concrete.

7. Precautions

- Keep the stirring speed, time, temperature and $BaCl_2$ quantity identical for samples and standards.
- Read within 4 min; the suspension settles/aggregates later.
- Concentrations above 40 mg/L must be diluted; $BaCl_2$ is toxic — avoid ingestion.

Experiment II.6: Determination of Optimum Coagulant Dosage by Jar Test

Course Outcome: CO4 | **Reference standard:** IS 3025 / CPHEEO Manual on Water Supply and Treatment (coagulation)

1. Aim

To determine the optimum dose of coagulant (alum) required for the efficient removal of turbidity from a raw water sample.

2. INPUTS — Apparatus, Materials & Quantities

Item	Specification / Quantity
Jar test apparatus	6-paddle flocculator with variable speed (0–100 rpm), six 1-litre beakers
Coagulant stock	1% alum solution: 10 g $Al_2(SO_4)_3 \cdot 18H_2O$ in 1 L (1 mL = 10 mg; 1 mL in 1 L = 10 mg/L)
Raw water	6 L of turbid water (initial turbidity 60 NTU, pH 7.6, alkalinity 80 mg/L as $CaCO_3$)
Instruments	Nephelometer, pH meter, stop-watch, 1/5/10 mL pipettes or syringes

3. Procedure

1. Fill the six beakers with 1 L of well-mixed raw water each and measure the initial turbidity, pH and alkalinity.
2. Add increasing alum doses (10, 20, 30, 40, 50, 60 mg/L, i.e., 1–6 mL of 1% stock) to the beakers simultaneously.
3. **Rapid mix**: 100 rpm for 1–2 min (coagulation).
4. **Slow mix**: 30 rpm for 15–20 min (flocculation); observe floc size.
5. Stop the paddles and let the water settle for 30 min; carefully withdraw supernatant from 2–3 cm below the surface.
6. Measure residual turbidity and pH of each jar; the dose giving the lowest turbidity with good floc is the **optimum dose**.

4. Observations — Jar test readings (raw water turbidity 60 NTU)

Sample readings are illustrative; students must replace them with their own laboratory readings.

Jar no.	Alum dose (mg/L)	Stock added (mL)	Floc formation	Residual turbidity (NTU)	Final pH	Removal (%)
1	10	1.0	Pin-point, poor	28	7.4	53.3
2	20	2.0	Small floc	12	7.3	80.0
3	30	3.0	Good floc	4.5	7.2	92.5
4	40	4.0	Large, fast-settling	2.5	7.1	95.8
5	50	5.0	Large, fast	3.0	7.0	95.0
6	60	6.0	Slightly turbid supernatant	4.2	6.9	93.0

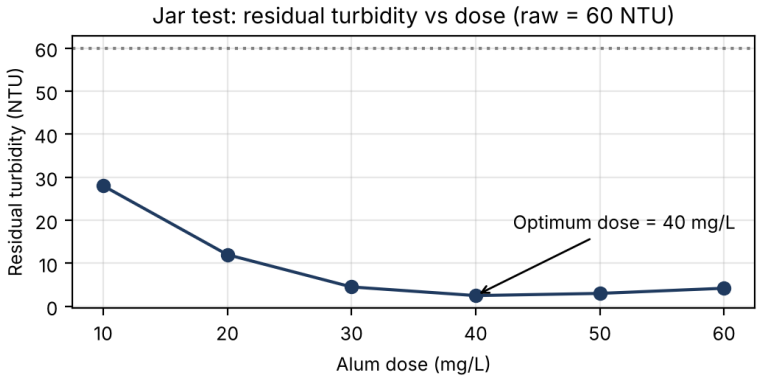


Fig. II.6 Residual turbidity against alum dose

5. Calculations

Volume of stock = Dose (mg/L) × Jar volume (L) ÷ Stock strength (10 mg/mL); 40 mg/L → 40 × 1 ÷ 10 = 4 mL

Turbidity removal (%) = (60 − 2.5) ÷ 60 × 100 = 95.8% at the optimum dose

Full-scale dose: for 10 MLD (10 000 m^3/d) at 40 g/m^3 → 40 × 10 000 = 400 000 g/d = 400 kg alum per day

Alkalinity consumed ≈ 0.5 mg/L as $CaCO_3$ per mg/L alum → 40 × 0.5 = 20 mg/L (available 80 mg/L – adequate)

6. OUTPUTS — Results from the sample data

Output parameter	Value	Remark
Optimum alum dose	40 mg/L	Lowest residual turbidity (2.5 NTU)
Turbidity removal	95.8 %	Final turbidity ≤ 5 NTU (IS 10500 permissible)
Alum for 10 MLD plant	400 kg/d	Rounded; add margin for dosing
Final pH	7.1	Within 6.5–8.5

Result / Acceptance criteria

Optimum alum dose = **40 mg/L**; doses above the optimum cause restabilisation (turbidity rises again at 50–60 mg/L).

7. Precautions

- Add coagulant to all jars simultaneously and mix at the stated speeds.
- Take supernatant without disturbing the settled floc.
- Rinse beakers and paddles well between tests.

Experiment II.7: Available Chlorine in Bleaching Powder and Residual Chlorine in Water

Course Outcome: CO2, CO4 | **Reference standard:** IS 1065 (bleaching powder); APHA 4500-Cl B (iodometric)

1. Aim

(a) To determine the percentage of available chlorine in bleaching powder; (b) to determine the residual chlorine in a disinfected water sample by the iodometric method.

2. INPUTS — Apparatus, Materials & Quantities

Item	Specification / Quantity
Bleaching powder	2.5 g, ground to a smooth paste with a little water and made up to 250 mL (10 g/L); 25 mL aliquot used
Sodium thiosulphate	0.1 N (24.82 g $Na_2S_2O_3 \cdot 5H_2O$/L) for (a); 0.0141 N for (b)
Reagents	10% KI solution; glacial acetic acid; starch indicator (1%)
Chlorinated water	500 mL sample (chlorine dose 1.0 mg/L, 30 min contact)
Glassware	Burette, 250/500 mL conical flasks, pipettes, volumetric flask

3. Procedure

1. **(a)** Pipette 25 mL of the bleaching-powder suspension into a conical flask, add 10 mL KI and 5 mL glacial acetic acid; liberated iodine turns the solution brown.
2. Titrate with 0.1 N $Na_2S_2O_3$ until the colour is pale yellow; add 1 mL starch (blue) and continue until the blue colour just disappears. Repeat for concordant titres.
3. **(b)** Take 500 mL of chlorinated water, add 5 mL acetic acid (pH 3–4) and about 1 g KI; mix.
4. Titrate with 0.0141 N $Na_2S_2O_3$ to a pale-yellow colour, add 1 mL starch and complete the titration to the disappearance of blue.
5. Run a blank (distilled water + reagents) and subtract any blank titre.

4. Observations — Titration readings

Sample readings are illustrative; students must replace them with their own laboratory readings.

(a) Bleaching powder	Trial 1	Trial 2	Mean
Aliquot (mL)	25	25	25
0.1 N $Na_2S_2O_3$ used (mL)	18.2	18.2	18.2

(b) Chlorinated water	Value
Sample volume (mL)	500
0.0141 N $Na_2S_2O_3$ used, A (mL)	0.4
Blank, B (mL)	0.0

5. Calculations

(a) 1 mL of 0.1 N $Na_2S_2O_3$ ≡ 3.545 mg Cl; Cl in aliquot = 18.2 × 0.1 × 35.45 = 64.5 mg

Mass of bleaching powder in 25 mL aliquot = 250 mg; Available chlorine (%) = 64.5 ÷ 250 × 100 = 25.8%

(b) Residual chlorine (mg/L) = (A − B) × N × 35 450 ÷ mL sample

= (0.4 − 0) × 0.0141 × 35 450 ÷ 500 = **0.40 mg/L**

Chlorine demand = Dose − Residual = 1.0 − 0.4 = 0.6 mg/L; Bleaching powder needed per 1000 m³ at 1 mg/L = 1 kg Cl_2 ÷ 0.258 = 3.9 kg

6. OUTPUTS — Results from the sample data

Output parameter	Value	Remark
Available chlorine in bleaching powder	25.8 %	Fresh powder 30–35%; should not be < 25% for use
Residual chlorine in water	0.40 mg/L	IS 10500: ≥ 0.2 mg/L (up to 1.0 mg/L permissible)
Chlorine demand	0.6 mg/L	Dose 1.0 mg/L

Result / Acceptance criteria

Bleaching powder contains **25.8% available chlorine** (slightly low — storage losses); residual chlorine of the treated water = **0.40 mg/L** (≥ 0.2 mg/L → adequate disinfection).

7. Precautions

- Titrate immediately after adding KI (iodine volatilises; protect from sunlight).
- Add starch only near the end point (pale yellow), otherwise iodine–starch complex is slow to decolourise.
- Store bleaching powder in a cool, dark, airtight container.

Experiment III.1: Estimation of Suspended, Volatile and Fixed Solids

Course Outcome: CO4 | **Reference standard:** IS 3025 (Part 15–18); APHA 2540 B, C, D, E

1. Aim

To determine the total, suspended, dissolved, volatile and fixed solids in a wastewater sample, as indicators of pollution load and sludge quantity.

2. INPUTS — Apparatus, Materials & Quantities

Item	Specification / Quantity
Sample	Domestic wastewater, well mixed; 50 mL for total solids, 100 mL for suspended solids
Evaporating dishes	Porcelain/platinum dishes (100 mL), pre-ignited at 550 °C, cooled in desiccator, weighed (W_1)
Filtration	Glass-fibre filter (1.5 µm, GF/C), filtration assembly with vacuum pump, Gooch crucibles
Ovens/furnace	Hot-air oven 103–105 °C; muffle furnace 550 ± 50 °C
Others	Desiccator, analytical balance (0.1 mg), measuring cylinder, tongs, steam bath

3. Procedure

1. **Total solids (TS):** pipette 50 mL well-mixed sample into the pre-weighed dish (W_1); evaporate on a steam bath; dry in oven at 103–105 °C for 1 h, cool in desiccator and weigh (W_2). Repeat drying until weight constant (±0.5 mg).
2. **Fixed and volatile solids:** ignite the same dish at 550 °C in the muffle furnace for 15–20 min; cool partially in air, then in a desiccator, and weigh (W_3). Loss on ignition = volatile solids.
3. **Total suspended solids (TSS):** wet and weigh a glass-fibre filter (F_1); filter 100 mL of well-mixed sample under suction; wash with distilled water; dry at 103–105 °C and weigh (F_2).
4. Ignite the filter at 550 °C for 15–20 min and weigh (F_3) to get volatile and fixed suspended solids.
5. **Dissolved solids:** TDS = TS − TSS (or filter through GF/C and evaporate the filtrate).

4. Observations — Weighings (all masses in g)

Sample readings are illustrative; students must replace them with their own laboratory readings.

Total solids (V = 50 mL)	W_1 dish (g)	W_2 after 105 °C (g)	W_3 after 550 °C (g)
Dish	45.231	45.267	45.246

Suspended solids (V = 100 mL)	F_1 filter (g)	F_2 after 105 °C (g)	F_3 after 550 °C (g)
Filter	0.112	0.134	0.127

5. Calculations

TS (mg/L) = $(W_2 - W_1) \times 10^6 \div V$ (mL) = $(45.2670 - 45.2310) \times 10^6 \div 50 = 36$ mg × 1000 ÷ 50 = 720 mg/L

VS (mg/L) = $(W_2 - W_3) \times 10^6 \div V = 21$ mg × 1000 ÷ 50 = 420 mg/L; FS = $(W_3 - W_1) \times 10^6 \div V = 15$ mg × 1000 ÷ 50 = 300 mg/L

TSS (mg/L) = $(F_2 - F_1) \times 10^6 \div V_f = 22$ mg × 1000 ÷ 100 = 220 mg/L; VSS = 15 × 10 = 150 mg/L; FSS = 7 × 10 = 70 mg/L

TDS = TS − TSS = 720 − 220 = 500 mg/L; VS/TS = 420 ÷ 720 = 58% (organic fraction)

6. OUTPUTS — Results from the sample data

Output parameter	Value	Remark
Total solids (TS)	720 mg/L	Medium-strength sewage 350–1200 mg/L
Volatile / Fixed solids	420 / 300 mg/L	VS/TS ≈ 58% – biodegradable
Total suspended solids (TSS)	220 mg/L	Typical raw sewage 100–350; discharge limit 100 mg/L
VSS / FSS	70 / 150 mg/L	—
Total dissolved solids (TDS)	500 mg/L	TS − TSS

Result / Acceptance criteria

Wastewater has TS = **720 mg/L**, TSS = **220 mg/L** (VSS 70), TDS = **500 mg/L**; the organic (volatile) fraction is about 58% — a medium-strength domestic sewage.

7. Precautions

- Cool dishes only in a desiccator and weigh promptly (hygroscopic).
- Do not exceed 550 °C (decomposition of carbonates); handle hot dishes with tongs.
- Mix sample well before pipetting; choose a volume giving 2.5–200 mg residue.

Experiment III.2: Determination of Sludge Volume Index (SVI) in Wastewater

Course Outcome: CO4, CO5 | **Reference standard:** APHA 2710 C/D (settled sludge volume, SVI)

1. Aim

To determine the Sludge Volume Index of activated sludge (mixed liquor) to evaluate its settling characteristics and detect sludge bulking.

2. INPUTS — Apparatus, Materials & Quantities

Item	Specification / Quantity
Mixed liquor	1 L fresh mixed-liquor sample from the aeration tank
Settling cylinder	1000 mL graduated measuring cylinder (≥ 25 cm height) with gentle stirrer
MLSS apparatus	Glass-fibre filter, vacuum filtration, oven at 103–105 °C, balance
Others	Stop-watch

3. Procedure

1. Pour 1000 mL of well-mixed mixed liquor into the settling cylinder; gently stir (1 rpm) to remove wall effects and start the stop-watch.
2. Record the sludge-blanket volume at 5, 10, 15, 20, 30, 45 and 60 min. The reading at **30 min (V_{30})** is used for SVI.
3. Simultaneously determine the MLSS of the same mixed liquor (filter 50 mL, dry at 103–105 °C, weigh).
4. Compute SVI and interpret the settleability.

4. Observations — Settling data and MLSS

Sample readings are illustrative; students must replace them with their own laboratory readings.

Time (min)	0	5	10	15	20	30	45	60
Sludge volume (mL/L)	1000	900	700	520	400	280	250	235

MLSS: filter before (g)	After 105 °C (g)	Sample (mL)	MLSS (mg/L)
0.112	0.272	50	3200

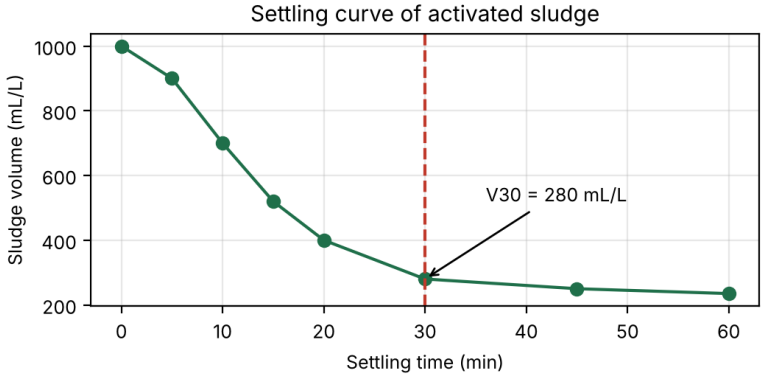


Fig. III.2 Settling curve of the sample

5. Calculations

MLSS = (0.2720 − 0.1120) × 10^6 ÷ 50 = 160 mg ÷ 0.05 L = 3200 mg/L

SVI (mL/g) = V_{30} (mL/L) × 1000 ÷ MLSS (mg/L) = 280 × 1000 ÷ 3200 = 87.5 mL/g

Sludge Density Index SDI = 100 ÷ SVI = 1.14 g/100 mL (good sludge SDI ≈ 1)

Interpretation: SVI < 100 mL/g – good settling; 100–150 – moderate; > 150 mL/g – bulking sludge

6. OUTPUTS — Results from the sample data

Output parameter	Value	Remark
MLSS	3200 mg/L	Conventional ASP 2000–4000 mg/L
Settled volume V_{30}	280 mL/L	—
Sludge Volume Index	87.5 mL/g	< 100 → good settling
Sludge Density Index	1.14	—

Result / Acceptance criteria

SVI = **87.5 mL/g** (< 100) — the activated sludge settles and compacts well, with no bulking problem.

7. Precautions

- Use fresh sample (analyse within 1 h); keep the cylinder away from vibration, direct sunlight and heat.
- Do not let the sludge adhere to cylinder walls; stir gently only once.
- Maintain temperature near that of the aeration tank.

Experiment III.3: Determination of Dissolved Oxygen (Winkler Azide-Modification)

Course Outcome: CO3 | **Reference standard:** IS 3025 (Part 38); APHA 4500-O C

1. Aim

To determine the dissolved oxygen (DO) of water and wastewater samples by the iodometric Winkler method (azide modification to avoid nitrite interference).

2. INPUTS — Apparatus, Materials & Quantities

Item	Specification / Quantity
Sample bottles	300 mL glass-stoppered BOD bottles, filled without air bubbles
$MnSO_4$ solution	480 g $MnSO_4 \cdot 4H_2O$ in 1 L distilled water
Alkali-iodide-azide	500 g NaOH + 135 g NaI (or 150 g KI) in 1 L; plus 10 g NaN_3 dissolved in 40 mL water
Conc. H_2SO_4	Sulphuric acid, sp. gr. 1.84
Titrant	0.025 N sodium thiosulphate (6.205 g $Na_2S_2O_3 \cdot 5H_2O$/L) – standardised
Indicator	Starch solution (1%)
Others	Burette, 250 mL conical flask, pipette, 200 mL measuring cylinder

3. Procedure

1. Collect the sample in a 300 mL BOD bottle by overflowing; add 2 mL $MnSO_4$ and 2 mL alkali-iodide-azide *below the liquid surface*; stopper (no air bubble) and mix by inverting 15–20 times.
2. A brown-orange precipitate appears (oxygen fixed). Allow it to settle to half the bottle height, then add 2 mL conc. H_2SO_4 and mix until the precipitate dissolves completely (yellow-brown iodine).
3. Measure 203 mL of the contents (≡ 200 mL of original sample, correcting for 4 mL of reagents) into a conical flask.
4. Titrate with 0.025 N $Na_2S_2O_3$ to a pale-straw colour; add 1–2 mL starch (blue) and continue to the first disappearance of blue.
5. Record the titre; repeat for concordance.

4. Observations — Titration readings (203 mL taken ≡ 200 mL sample; temperature 27 °C)

Sample readings are illustrative; students must replace them with their own laboratory readings.

Sample	Burette initial (mL)	Burette final (mL)	Titre V (mL)
River water	0.0	6.2	6.2
Raw sewage	0.0	0.4	0.4
Treated effluent	0.0	4.8	4.8

5. Calculations

DO (mg/L) = [V × N × 8 × 1000] ÷ V_s (mL sample titrated, corrected)

With 0.025 N titrant and 200 mL sample: DO (mg/L) = V × 0.025 × 8 × 1000 ÷ 200 = V (1 mL titrant ≡ 1 mg/L DO)

River water: DO = 6.2 mg/L; % saturation = DO ÷ DO_{sat} × 100 = 6.2 ÷ 8.0 × 100 = 77.5% (DO_{sat} at 27 °C ≈ 8.0 mg/L)

6. OUTPUTS — Results from the sample data

Output parameter	Value	Remark
River water	DO = 6.2 mg/L (77.5% sat.)	≥ 5 mg/L for aquatic life / Class A–C waters (CPCB)
Raw sewage	DO = 0.4 mg/L (5% sat.)	Septic/anaerobic tendency
Treated effluent	DO = 4.8 mg/L (60% sat.)	Aerobic treatment effective

Result / Acceptance criteria

DO of river water = **6.2 mg/L**, raw sewage = **0.4 mg/L** and treated effluent = **4.8 mg/L**. Higher DO indicates cleaner, better-aerated water; DO is the basis for the BOD test (Exp. III.4).

7. Precautions

- Avoid air entrapment while filling the bottle and while adding reagents.
- Fix the sample in the field immediately; analyse within 8 h, in the dark.
- Add starch near the end point only; handle conc. H_2SO_4 and sodium azide (toxic, explosive with acids/metals) carefully.

Experiment III.4: Estimation of B.O.D. (5-day, 20 °C)

Course Outcome: CO3 | **Reference standard:** IS 3025 (Part 44); APHA 5210 B

1. Aim

To determine the biochemical oxygen demand, the amount of oxygen needed by microorganisms to stabilise the biodegradable organic matter in a wastewater sample in 5 days at 20 °C.

2. INPUTS — Apparatus, Materials & Quantities

Item	Specification / Quantity
BOD bottles	300 mL glass-stoppered bottles (6 for 3 dilutions × initial/final)
Incubator	20 ± 1 °C, dark
Dilution water	Distilled water aerated to DO ≈ 8–9 mg/L; per litre add 1 mL each of phosphate buffer (pH 7.2), $MgSO_4$ (22.5 g/L), $CaCl_2$ (27.5 g/L), $FeCl_3$ (0.25 g/L)
Seed	1–2 mL settled domestic sewage per litre (if sample is unseeded/industrial)
DO reagents	As in Exp. III.3 ($MnSO_4$, alkali-iodide-azide, H_2SO_4, 0.025 N thiosulphate, starch)
Sample	Domestic wastewater (neutral pH, free of chlorine), 500 mL
Others	Pipettes, measuring cylinders, water seal for bottles

3. Procedure

1. Prepare dilutions (1%, 2%, 3%) by adding 3, 6 and 9 mL of sample to separate BOD bottles and filling to the neck with dilution water (siphon, no bubbles); also keep a dilution-water blank.
2. Determine the initial DO (D_1) of one set of bottles by the Winkler method immediately after preparation.
3. Stopper the duplicate bottles, add a water seal, and incubate at 20 °C in the dark for 5 days.
4. After 5 days determine the final DO (D_2) of each bottle.
5. Select dilutions giving a DO depletion ≥ 2 mg/L and residual DO ≥ 1 mg/L; calculate BOD for those and average.

4. Observations — DO readings

Sample readings are illustrative; students must replace them with their own laboratory readings.

Bottle	Sample (mL in 300 mL)	Dilution factor P	Initial DO D_1 (mg/L)	5-day DO D_2 (mg/L)	Depletion $D_1 - D_2$	Valid?	BOD_5 (mg/L)
A	3	0.01	8.2	5.7	2.5	Yes	250
B	6	0.02	8.1	3.2	4.9	Yes	245
C	9	0.03	8.0	0.6	7.4	No (DO < 1)	—
Blank	0	—	8.3	8.2	0.1	≤ 0.2 ✓	—

5. Calculations

$BOD\ (mg/L) = (D_1 - D_2) \div P$, where P = Volume of sample ÷ Volume of bottle = 3 ÷ 300 = 0.01

Bottle A: BOD = (8.2 − 5.7) ÷ 0.01 = 250 mg/L; Bottle B: (8.1 − 3.2) ÷ 0.02 = 245 mg/L; Bottle C rejected (residual DO 0.6 < 1 mg/L)

Mean BOD_5 = (250 + 245) ÷ 2 = 247.5 mg/L

Ultimate BOD (first-order, k = 0.23 d^{-1} at 20 °C): $L_0 = BOD_5 \div (1 - e^{-5k}) = 247.5 \div 0.683 = 362$ mg/L

BOD_3 at 27 °C (Indian standard) ≈ 0.9 × BOD_5 at 20 °C

6. OUTPUTS — Results from the sample data

Output parameter	Value	Remark
BOD_5 at 20 °C	247.5 mg/L	Medium-strength sewage 200–300 mg/L
Ultimate BOD, L_0	362 mg/L	≈ 1.46 × BOD_5
Blank depletion	0.1 mg/L	Must be ≤ 0.2 mg/L
Compliance (30 mg/L)	Does not comply	Treatment required before discharge

Result / Acceptance criteria

BOD_5 of the wastewater = **247.5 mg/L** (about 8.2× the 30 mg/L discharge limit of CPCB) — the sewage is of medium strength and needs biological treatment.

7. Precautions

- Use only dilutions that satisfy depletion ≥ 2 mg/L and residual DO ≥ 1 mg/L.
- Do not allow air bubbles; keep the water seal full; incubate in the dark (algal photosynthesis).
- Neutralise samples to pH 6.5–7.5; dechlorinate with sodium sulphite if chlorinated.

Experiment III.5: Estimation of C.O.D. (Open Reflux Dichromate Method)

Course Outcome: CO3 | **Reference standard:** IS 3025 (Part 58); APHA 5220 B

1. Aim

To determine the chemical oxygen demand — the oxygen equivalent of organic matter oxidised by a strong chemical oxidant (potassium dichromate) — of a wastewater sample.

2. INPUTS — Apparatus, Materials & Quantities

Item	Specification / Quantity
Reflux apparatus	250/500 mL round-bottom flasks with ground-glass joints, condensers, heating mantle (open reflux, 2 h)
$K_2Cr_2O_7$ 0.25 N	12.259 g $K_2Cr_2O_7$ (dried at 103 °C) in 1 L distilled water
Sulphuric acid reagent	Ag_2SO_4 (5.5 g/kg) in conc. H_2SO_4 (catalyst)
Mercuric sulphate	$HgSO_4$ crystals (≈ 0.4 g per 20 mL sample) to complex chloride
Titrant	0.1 N ferrous ammonium sulphate (FAS: 39.2 g $Fe(NH_4)_2(SO_4)_2{\cdot}6H_2O$ + 20 mL conc. H_2SO_4 per L), standardised against $K_2Cr_2O_7$
Indicator	Ferroin (1,10-phenanthroline-ferrous sulphate), 2–3 drops
Others	Glass beads, burette, pipettes, sample 20 mL (diluted if necessary)

3. Procedure

1. Place 20 mL of sample (or an aliquot diluted to 20 mL) in the reflux flask; add 0.4 g $HgSO_4$ and a few glass beads.
2. Slowly add 10 mL 0.25 N $K_2Cr_2O_7$ and mix; attach the condenser and add 30 mL Ag_2SO_4-H_2SO_4 reagent through the top, swirling carefully while cooling.
3. Reflux for 2 h on the heating mantle; cool and rinse the condenser with distilled water; dilute to about 140 mL.
4. Add 2–3 drops of ferroin and titrate the excess dichromate with 0.1 N FAS to the colour change *blue-green → reddish-brown*.
5. Run a blank with 20 mL distilled water and the same reagents.

4. Observations — Titration readings (FAS 0.1 N)

Sample readings are illustrative; students must replace them with their own laboratory readings.

Experiment	Sample vol. (mL)	FAS used (mL)
Blank (A)	20	25.0
Sample, trial 1	20	14.6
Sample, trial 2	20	14.8
Mean sample (B)	20	14.7

5. Calculations

COD (mg O_2/L) = (A − B) × N × 8000 ÷ V_{sample} (A = FAS for blank, B = FAS for sample)

= (25.0 − 14.7) × 0.1 × 8000 ÷ 20 = 10.3 × 40 = **412 mg/L**

Biodegradability ratio BOD_5/COD = 247 ÷ 412 = 0.60

Check: 1 g potassium hydrogen phthalate (KHP) = 1.176 g COD (use as verification standard)

6. OUTPUTS — Results from the sample data

Output parameter	Value	Remark
COD	412 mg/L	Medium-strength sewage 400–500 mg/L
BOD_5 / COD	0.60	> 0.5 → readily biodegradable (biological treatment suitable)
Compliance (250 mg/L)	Does not comply	CPCB inland-surface-water limit 250 mg/L

Result / Acceptance criteria

COD of the wastewater = **412 mg/L**; BOD_5/COD = 0.60, so the waste is largely biodegradable and suited to biological treatment.

7. Precautions

- Dichromate is a Cr(VI) carcinogen and Hg/Ag wastes are toxic — collect all residues in a designated waste container.
- Add acid slowly with mixing to avoid hot spots/ejection; use glass beads to prevent bumping.
- Standardise FAS daily; use clean, grease-free glassware.

Experiment III.6: Determination of TKN and Ammonia Nitrogen in Wastewater

Course Outcome: CO3 | **Reference standard:** IS 3025 (Part 34); APHA 4500-N_{org} B (macro-Kjeldahl), 4500-NH_3 B/C

1. Aim

To determine ammonia nitrogen (NH_3-N) and total Kjeldahl nitrogen (TKN = organic N + ammonia-N) of wastewater by distillation and titration.

2. INPUTS — Apparatus, Materials & Quantities

Item	Specification / Quantity
Kjeldahl apparatus	800 mL Kjeldahl flasks, digestion heater/block with fume extraction, distillation assembly with condenser
Digestion reagent	134 g K_2SO_4 + 7.3 g $CuSO_4$ + 134 mL conc. H_2SO_4 in 1 L water
Hydroxide–thiosulphate	500 g NaOH + 25 g $Na_2S_2O_3 \cdot 5H_2O$ per litre (for alkaline distillation)
Borate buffer (pH 9.5)	88 mL 0.1 N NaOH + 500 mL 0.025 M $Na_2B_4O_7$ diluted to 1 L (for NH_3-N distillation)
Receiving solution	2% boric acid with mixed indicator (methyl red + bromocresol green)
Titrant	0.02 N H_2SO_4 (standardised)
Sample	50 mL wastewater (settled, pH adjusted to 7)

3. Procedure

1. **Ammonia-N:** take 50 mL sample + 25 mL borate buffer (adjust to pH 9.5 with NaOH); distil about 40 mL into 50 mL boric acid + indicator (condenser tip below liquid).
2. Titrate the distillate (green → pink/violet) with 0.02 N H_2SO_4; run a blank with distilled water.
3. **TKN:** take 50 mL fresh sample, add 10 mL digestion reagent and a few glass beads; boil off water until white SO_3 fumes appear and digest for 30 min more until the solution is clear/pale green.
4. Cool, dilute to 300 mL, add 50 mL hydroxide–thiosulphate (turns brown/blue-black); distil and collect the ammonia in boric acid; titrate with 0.02 N H_2SO_4; run a digestion blank.
5. Organic-N = TKN − NH_3-N.

4. Observations — Titration readings (0.02 N H_2SO_4, sample 50 mL)

Sample readings are illustrative; students must replace them with their own laboratory readings.

Determination	Sample (mL)	Titre – sample A (mL)	Titre – blank B (mL)	(A − B) (mL)
Ammonia-N	50	4.2	0.1	4.1
TKN	50	7.3	0.1	7.2

5. Calculations

N (mg/L) = (A − B) × N_{acid} × 14 000 ÷ V_{sample} (1 mL of 0.02 N acid ≡ 0.28 mg N)

Ammonia-N = (4.2 − 0.1) × 0.02 × 14 000 ÷ 50 = 4.1 × 5.6 = 23.0 mg/L

TKN = (7.3 − 0.1) × 0.02 × 14 000 ÷ 50 = 7.2 × 5.6 = 40.3 mg/L

Organic-N = TKN − NH_3-N = 40.3 − 23.0 = 17.4 mg/L

6. OUTPUTS — Results from the sample data

Output parameter	Value	Remark
Ammonia-N	23.0 mg/L	CPCB limit 50 mg/L ✓
TKN	40.3 mg/L	CPCB limit 100 mg/L ✓; medium sewage ≈ 40 mg/L
Organic nitrogen	17.4 mg/L	≈ 43% of TKN

Result / Acceptance criteria

Ammonia-N = **23.0 mg/L**, TKN = **40.3 mg/L** and organic-N = **17.4 mg/L**; BOD_5 : N ≈ 6.1 : 1, i.e., sufficient nitrogen for biological treatment.

7. Precautions

- Digest in a fume hood; avoid ammonia contamination from the laboratory air (do not store near NH_4OH).

- Make sure the delivery tube dips into the receiving boric acid and that there is no suck-back.
- NaOH solution is strongly corrosive; boiling H_2SO_4 gives dense fumes — wear face shield and gloves.

Experiment III.7: Determination of Total and Faecal Coliform (Demonstration Only)

Course Outcome: CO5 | **Reference standard:** IS 1622; APHA 9221 B, C, E (Multiple-tube fermentation, MPN)

1. Aim

To detect and enumerate total coliform and faecal (thermotolerant) coliform bacteria in water/wastewater by the Multiple-Tube Fermentation (MPN) technique as indicator of faecal pollution.

2. INPUTS — Apparatus, Materials & Quantities

Item	Specification / Quantity
Media	Lauryl tryptose broth (presumptive), Brilliant green lactose bile broth (BGLB, total coliform), EC broth (faecal coliform) with inverted Durham tubes
Dilution water	Phosphate-buffered/peptone water, sterile (9 mL blanks)
Equipment	Autoclave (121 °C, 15 min), incubators at 35 ± 0.5 °C and water bath 44.5 ± 0.2 °C, laminar-flow hood
Glassware	15 test tubes (5 × 10 mL, 5 × 1 mL, 5 × 0.1 mL) with Durham tubes, sterile pipettes, inoculation loop, Bunsen burner
Sample	Collected in a sterile bottle, dechlorinated with $Na_2S_2O_3$, kept at < 10 °C, analysed within 6 h

3. Procedure

1. **Presumptive test:** inoculate 5 tubes of double-strength broth with 10 mL, 5 tubes of single-strength broth with 1 mL, and 5 tubes with 0.1 mL of sample; incubate at 35 °C for 24 ± 2 h and then 48 ± 3 h.
2. Note tubes showing gas (≥ 10% of Durham tube) and acid (yellow colour) — presumptive positive.
3. **Confirmed total coliform:** transfer a loopful from every positive tube to BGLB broth; incubate at 35 °C for 48 h; gas production confirms total coliform.
4. **Faecal coliform:** transfer from positive presumptive tubes to EC broth; incubate in a water bath at 44.5 °C for 24 h; gas production confirms faecal (thermotolerant) coliform.
5. Count the number of positive tubes in each dilution series and read the MPN index/100 mL from the standard MPN table.

4. Observations — Tube results (5-tube, 10 / 1 / 0.1 mL series)

Sample readings are illustrative; students must replace them with their own laboratory readings.

Test	Medium / temp.	Positive tubes: 10 mL	1 mL	0.1 mL	Combination	MPN / 100 mL
Presumptive	Lauryl tryptose, 35 °C	5	4	2	5-4-2	—
Total coliform (confirmed)	BGLB, 35 °C	5	3	1	5-3-1	110
Faecal coliform	EC broth, 44.5 °C	5	2	0	5-2-0	49

5. Calculations

MPN/100 mL is read from the MPN table for the combination of positive tubes (10-1-0.1 mL): 5-3-1 → 110; 5-2-0 → 49

If dilutions other than 10-1-0.1 mL are used: MPN/100 mL = MPN index × (10 ÷ largest volume of the selected dilutions in mL) — e.g., for 1-0.1-0.01 mL series multiply by 10

Faecal fraction = 49 ÷ 110 = 45% of coliforms are thermotolerant (faecal origin)

6. OUTPUTS — Results from the sample data

Output parameter	Value	Remark
Total coliform	110 MPN/100 mL	Drinking water (IS 10500): absent in 100 mL — **not potable**
Faecal coliform	49 MPN/100 mL	Indicates recent faecal contamination
Treated-sewage discharge	—	Faecal coliform norm < 1000 MPN/100 mL (NGT/CPCB STP norm) – meets

Result / Acceptance criteria

The water sample contains **110 MPN/100 mL total coliform** and **49 MPN/100 mL faecal coliform**; it is faecally contaminated and must be disinfected (chlorination) before drinking.

7. Precautions

- Use aseptic technique; sterilise loops by flaming; wear lab coat and gloves.
- Decontaminate cultures by autoclaving before disposal; never pipette by mouth.
- Incubate for the exact time and temperature (44.5 ± 0.2 °C for faecal).

Appendix A: Summary of Experiments, Standards, Safety and Books

No.	Experiment	Key input	Sample output	Limit / reference
II.1	Sampling & preservation	2 L samples, preservatives, 4 °C	Composite 2000 mL	Holding times (APHA)
II.2	EC & turbidity	0.01 M KCl; formazin	373–981 µS/cm; 0.8–18.6 NTU	Turbidity 1/5 NTU; TDS 500/2000
II.3	Fluoride	SPADNS; 570 nm	0.51 mg/L	1.0 / 1.5 mg/L
II.4	Iron	Phenanthroline; 510 nm	0.26 mg/L	0.3 mg/L
II.5	Sulphate	$BaCl_2$ turbidimetry; 420 nm	90 mg/L	200 / 400 mg/L
II.6	Jar test	1% alum; 6 jars	40 mg/L optimum; 95.8% removal	Turbidity ≤ 5 NTU
II.7	Chlorine	0.1 N / 0.0141 N thiosulphate	25.8% available Cl; 0.40 mg/L residual	Residual ≥ 0.2 mg/L
III.1	Solids	50 mL / 100 mL	TS 720, TSS 220, VS 420 mg/L	TSS ≤ 100 mg/L
III.2	SVI	1 L mixed liquor	SVI 87.5 mL/g	< 100 good; > 150 bulking
III.3	DO	300 mL; 0.025 N thiosulphate	6.2 / 0.4 / 4.8 mg/L	≥ 5 mg/L (surface water)
III.4	BOD_5	3 dilutions; 20 °C, 5 d	247.5 mg/L	30 mg/L (3 d, 27 °C)
III.5	COD	20 mL; 0.1 N FAS	412 mg/L	250 mg/L
III.6	TKN / NH_3-N	50 mL; 0.02 N H_2SO_4	40.3 / 23.0 mg/L	TKN 100; NH_4-N 50 mg/L
III.7	Coliform (MPN)	5-tube, 15 tubes	110 / 49 MPN/100 mL	Absent in 100 mL (drinking)

General laboratory safety

- Wear lab coat, gloves and goggles; never pipette by mouth; work with concentrated acids, dichromate, azide, mercury and silver salts in a fume hood.
- Collect Cr(VI), Hg and Ag wastes separately in labelled containers; do not pour them down the drain.
- Treat all wastewater and bacteriological samples as infectious; autoclave cultures before disposal; wash hands after work.
- Know the location of the eyewash, safety shower, first-aid kit and fire extinguisher.
- Calibrate pH meters, balances and spectrophotometers; standardise titrants (thiosulphate, FAS, H_2SO_4) before use.

Text books

1. Manual on Water Supply and Treatment, Ministry of Urban Development, New Delhi.
2. Manual on Sewerage and Sewage Treatment Systems, Part A, B and C, Central Public Health and Environmental Engineering Organization (CPHEEO), Ministry of Urban Development.

Reference book

1. Environmental Engineering Laboratory Manual by Dr. S.K. Panigrahi, L. Mohanty, S.K. Kataria & Sons.

Online learning resources

- https://ee1-nitk.vlabs.ac.in/
- https://ee2-nitk.vlabs.ac.in/

Note: limits are from IS 10500:2012, CPCB / Environment (Protection) Rules, 1986 General Standards and APHA Standard Methods; confirm with the latest editions. All observation data in this manual are illustrative and must be replaced with actual laboratory readings.